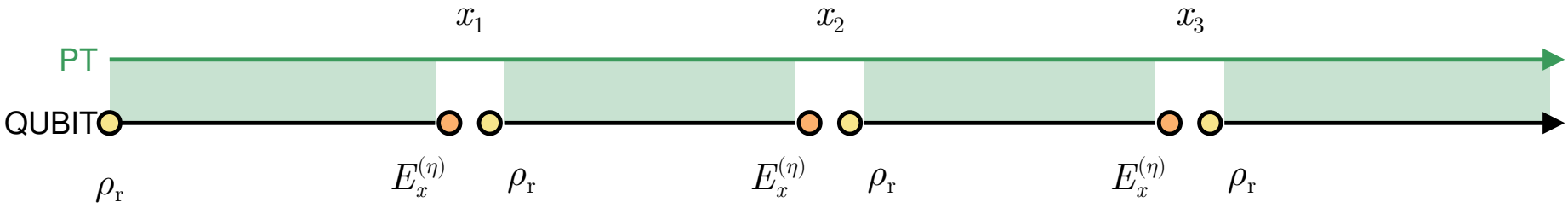
Three repeated Ramsey readouts

$$\rho_{\mathrm{r}} = |+\rangle\langle+| \qquad E_x^{(\eta)} = \frac{1}{2}(I + x\eta\sigma_x),\ \eta = 0.90$$